ECE 555 Project — Milestone 1: Gate-Level Design

Single MLP neuron, ASAP7 (7 nm FinFET), Cadence Virtuoso - Fall 2026

Team: Arihan Yadav | Abhinav Nandwani | _____

1. What the neuron computes

The neuron computes **z = ReLU(y)**, where **$y = x_0 \cdot w_0 + x_1 \cdot w_1 + b$**. Inputs, weights and the bias weight are 2-bit two's-complement values restricted to {−1, 0, +1}, coded as `11`, `00` and `01`. The code `10` is never applied. That gives $-3 \le y \le +3$, so $z \in \{0, 1, 2, 3\}$ and the output is the 3-bit word Z2 Z1 Z0.

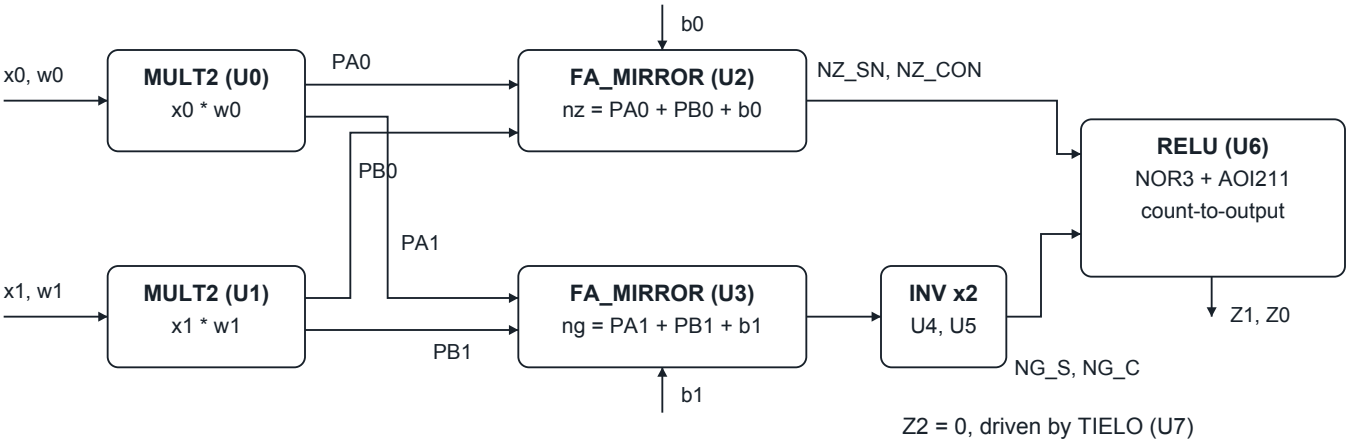
2. Architecture

Every term t (the two products and the bias) is itself in {−1, 0, +1}, and its two bits have a direct meaning. **$t_0 = 1$ means the term is nonzero**, and **$t_1 = 1$ means the term is negative**. If nz is the number of nonzero terms and ng is the number of negative terms, then

$$\mathbf{y = (\#positive) - (\#negative) = nz - 2 \cdot ng}, \quad z = \max(0, nz - 2 \cdot ng).$$

So the datapath is (1) two **2-bit multipliers**, (2) one **full adder counting the three bit-0s** (nz) and one **full adder counting the three bit-1s** (ng), and (3) a small **ReLU logic block** that turns the two 2-bit counts into z.

This is the spec's multiplier → 2-bit adder → 3-bit adder → 3-bit 2:1 MUX chain in carry-save form. The two full adders add all three operands in one level with no carry ripple between them. The ReLU block merges the sign-selected MUX with the conversion from (nz, ng) to binary. The slides note that the example blocks are optional, and we chose this form because it has the shortest logic depth and the fewest transistors.



2.1 Truth tables

Multiplier truth table (legal codes)

x	x_1x_0	w	w_1w_0	p	p_1p_0
-1	11	-1	11	+1	01
-1	11	+0	00	+0	00
-1	11	+1	01	-1	11
+0	00	-1	11	+0	00
+0	00	+0	00	+0	00
+0	00	+1	01	+0	00
+1	01	-1	11	-1	11
+1	01	+0	00	+0	00
+1	01	+1	01	+1	01

$P0 = x_0 \cdot w_0 \quad P1 = x_0 \cdot w_0 \cdot (x_1 \oplus w_1)$

ReLU logic (all reachable counts, ng ≤ nz)

nz	c s	ng	c s	y	z	Z2Z1Z0
0	00	0	00	+0	0	000
1	01	0	00	+1	1	001
1	01	1	01	-1	0	000
2	10	0	00	+2	2	010
2	10	1	01	+0	0	000
2	10	2	10	-2	0	000
3	11	0	00	+3	3	011
3	11	1	01	+1	1	001
3	11	2	10	-1	0	000
3	11	3	11	-3	0	000

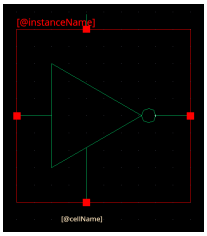
$Z1 = c_{nz} \cdot \bar{s}_{ng} \cdot \bar{c}_{ng} \quad Z0 = s_{nz} \cdot \bar{c}_{ng} \cdot (c_{nz} + \bar{s}_{ng}) \quad Z2 = 0$

3. Sub-module schematics

Each block is its own cell in library `NEURON`, with a schematic and a symbol, and higher levels are built only from those symbols. The leaf gates are static CMOS using ASAP7 RVT devices (`nmos_rvt` / `pmos_rvt`, L = 20 nm). VDD and VSS are explicit inputOutput pins rather than global nets, as Tutorial 2 recommends for LVS. The initial sizing is **2 fins per NMOS and 3 fins per PMOS**. Sizing will be refined in Milestones 2 and 3 against the (f_{max}/area) metric. Net names are shown as wire labels.

3.1 INV

Inverter: Y = NOT A. 2 transistors.

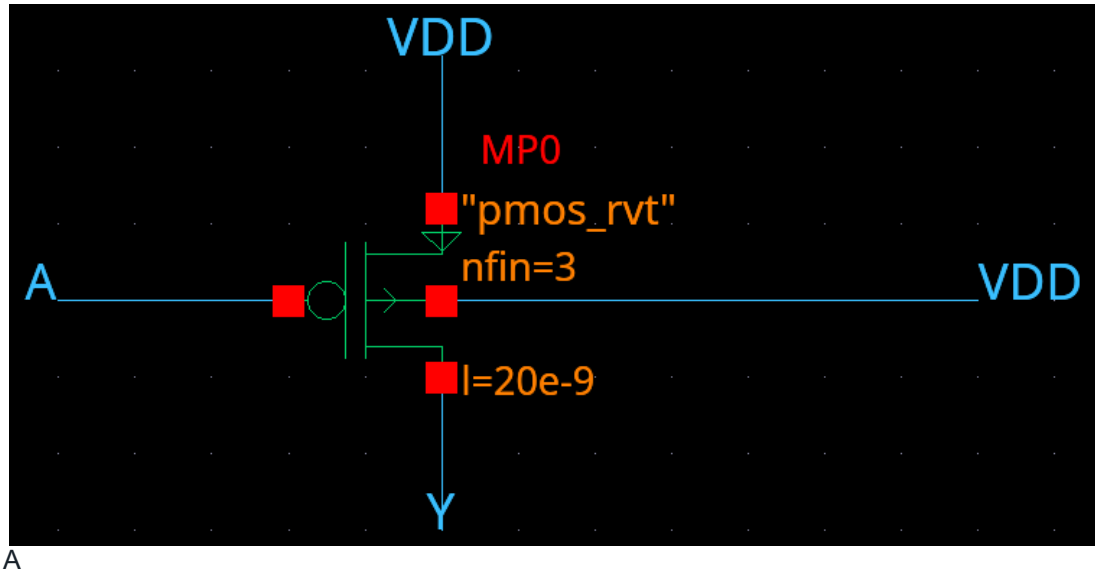


Cadence cell symbol

Inputs: A
Outputs: Y
Supplies: VDD, VSS

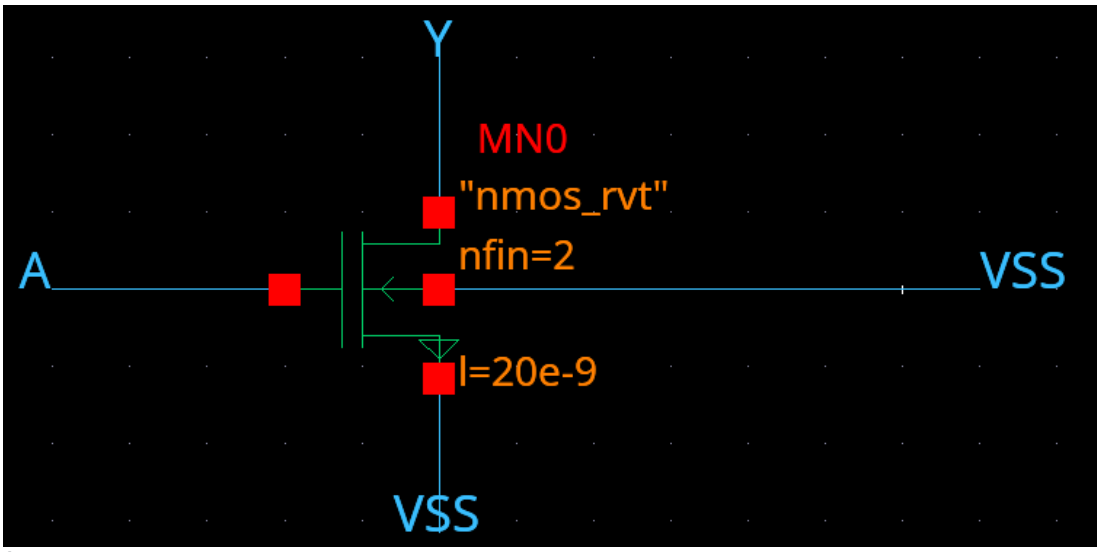
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MP0 | PMOS | 3 fins



D = Y G = A
S = VDD Bulk = VDD

MN0 | NMOS | 2 fins

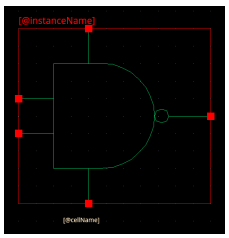


D = Y G = A
S = VSS Bulk = VSS

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

3.2 NAND2

Two-input NAND: Y = NOT (A AND B). 4 transistors.

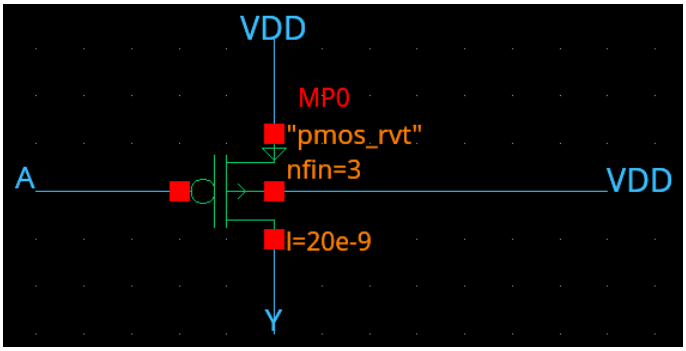


Cadence cell symbol

Inputs: A, B
Outputs: Y
Supplies: VDD, VSS

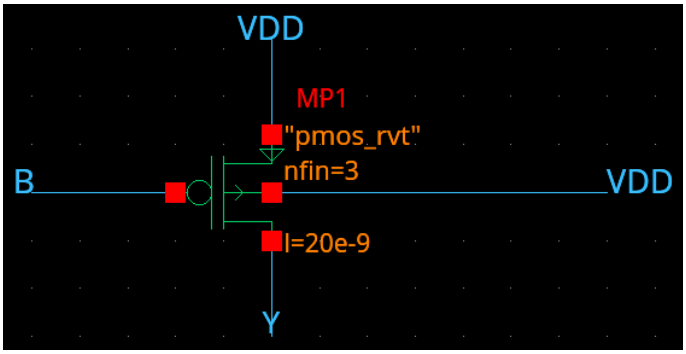
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MP0 | PMOS | 3 fins



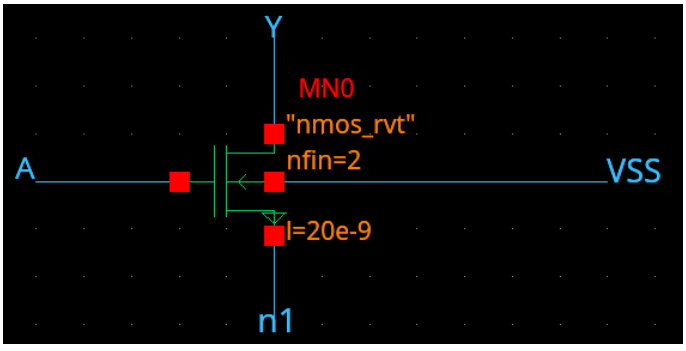
D = Y G = A
S = VDD Bulk = VDD

MP1 | PMOS | 3 fins



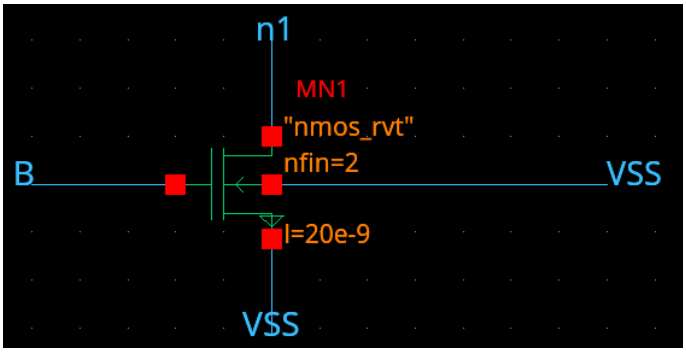
D = Y G = B
S = VDD Bulk = VDD

MN0 | NMOS | 2 fins



D = Y G = A
S = n1 Bulk = VSS

MN1 | NMOS | 2 fins

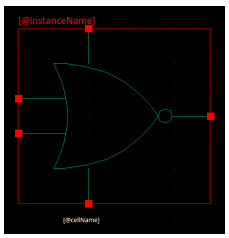


D = n1 G = B
S = VSS Bulk = VSS

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

3.3 NOR2

Two-input NOR: Y = NOT (A OR B). 4 transistors.

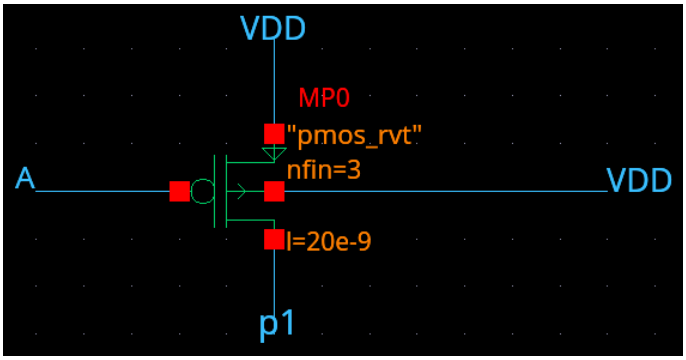


Cadence cell symbol

Inputs: A, B
Outputs: Y
Supplies: VDD, VSS

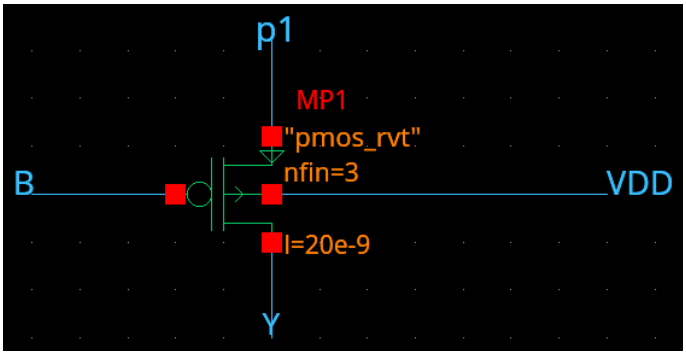
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MP0 | PMOS | 3 fins



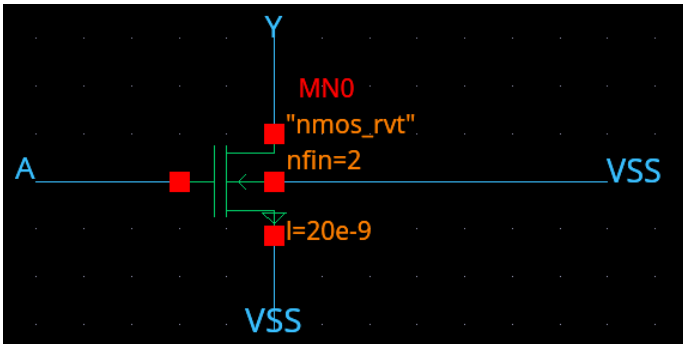
D = p1 G = A
S = VDD Bulk = VDD

MP1 | PMOS | 3 fins



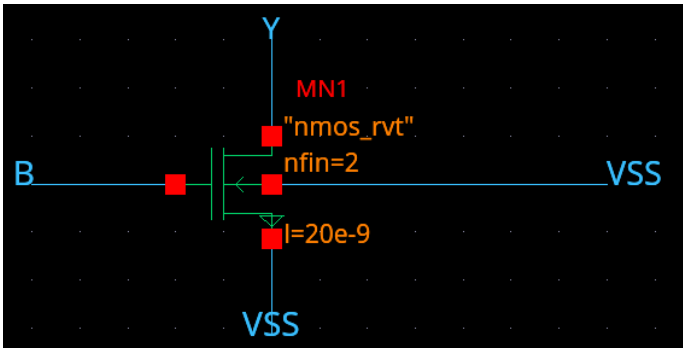
D = Y G = B
S = p1 Bulk = VDD

MN0 | NMOS | 2 fins



D = Y G = A
S = VSS Bulk = VSS

MN1 | NMOS | 2 fins



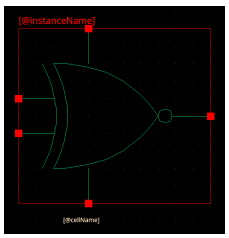
D = Y G = B
S = VSS Bulk = VSS

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

3.4 XNOR2 - details 1/2

Static CMOS XNOR: Y = A*B + NOT(A)*NOT(B). 12 transistors; input inverters and an AOI22 core.

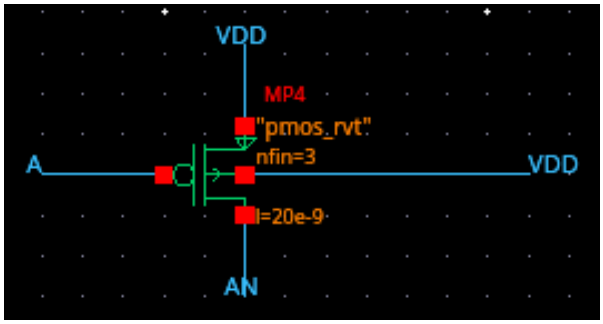
Inputs: A, B
Outputs: Y
Supplies: VDD, VSS



Cadence cell symbol

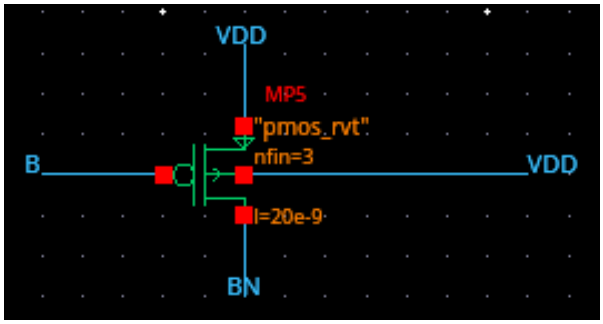
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MP4 | PMOS | 3 fins



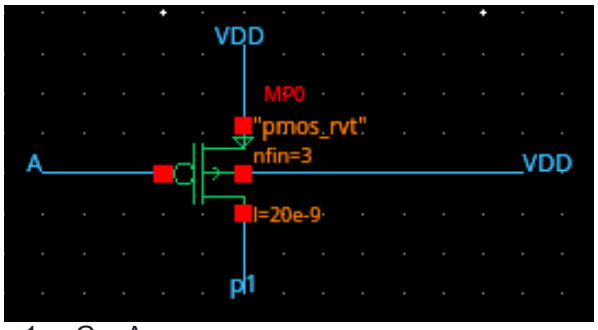
D = AN G = A
S = VDD Bulk = VDD

MP5 | PMOS | 3 fins



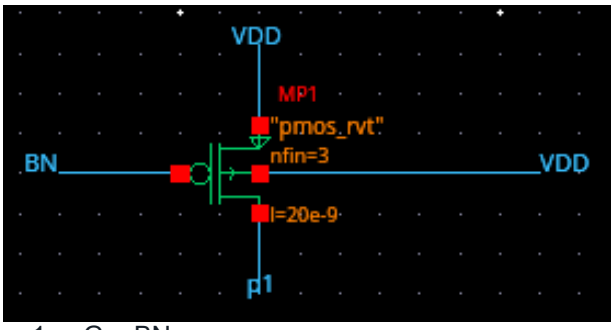
D = BN G = B
S = VDD Bulk = VDD

MP0 | PMOS | 3 fins



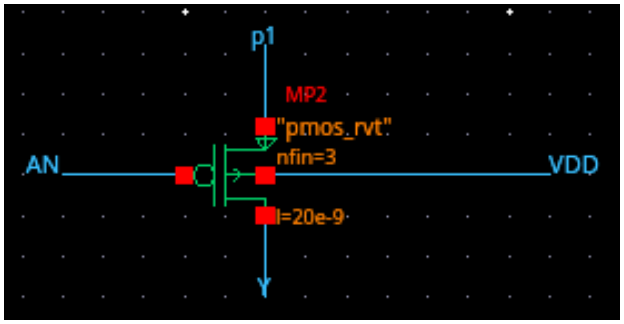
D = p1 G = A
S = VDD Bulk = VDD

MP1 | PMOS | 3 fins



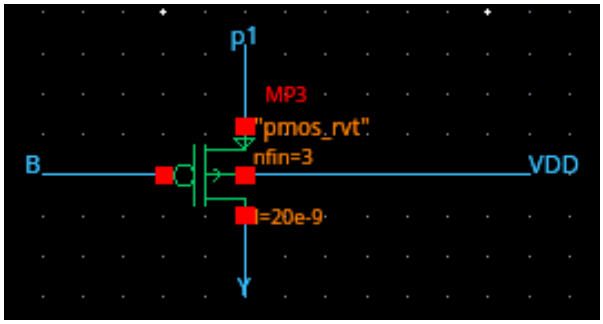
D = p1 G = BN
S = VDD Bulk = VDD

MP2 | PMOS | 3 fins



D = Y G = AN
S = p1 Bulk = VDD

MP3 | PMOS | 3 fins



D = Y G = B
S = p1 Bulk = VDD

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

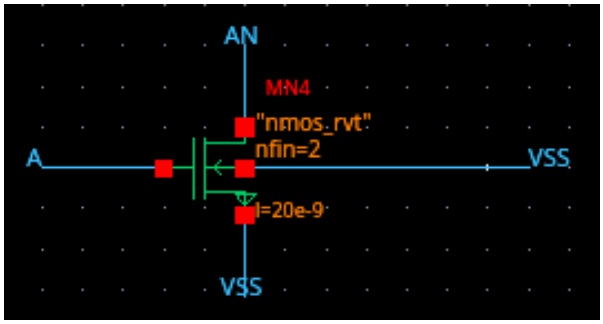
3.4 XNOR2 - details 2/2

Static CMOS XNOR: Y = A*B + NOT(A)*NOT(B). 12 transistors; input inverters and an AOI22 core.

Inputs: A, B
Outputs: Y
Supplies: VDD, VSS

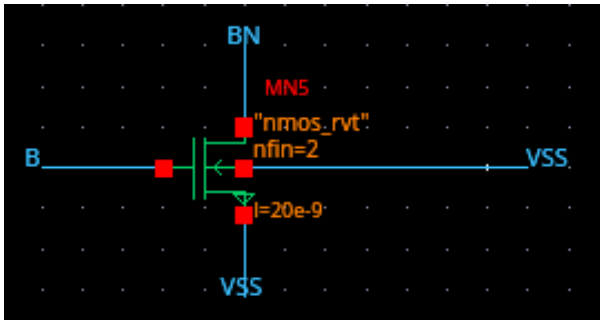
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MN4 | NMOS | 2 fins



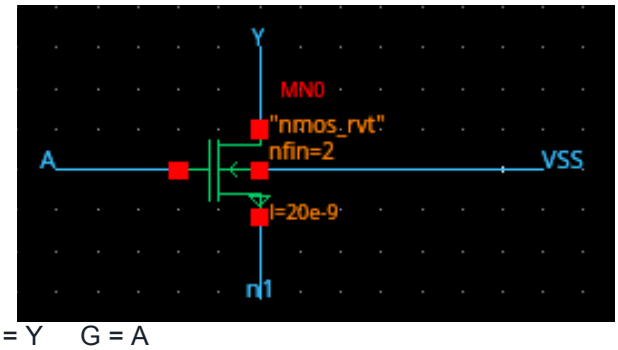
D = AN G = A
S = VSS Bulk = VSS

MN5 | NMOS | 2 fins



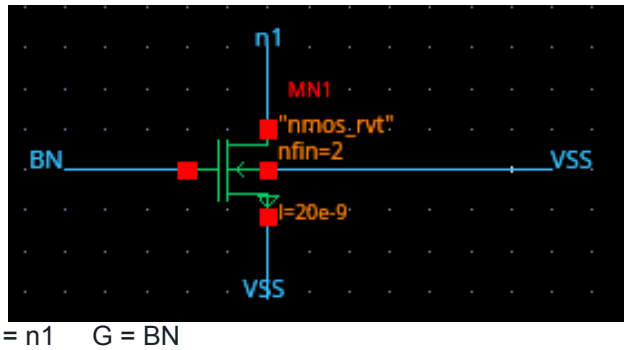
D = BN G = B
S = VSS Bulk = VSS

MN0 | NMOS | 2 fins



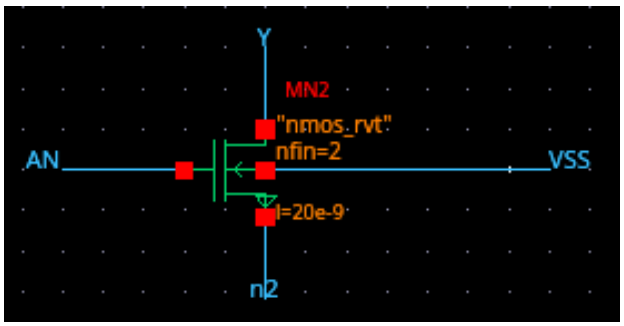
D = Y G = A
S = n1 Bulk = VSS

MN1 | NMOS | 2 fins



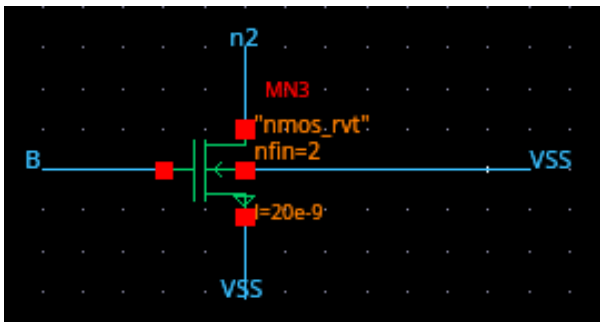
D = n1 G = BN
S = VSS Bulk = VSS

MN2 | NMOS | 2 fins



D = Y G = AN
S = n2 Bulk = VSS

MN3 | NMOS | 2 fins



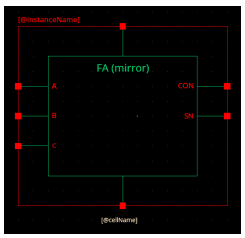
D = n2 G = B
S = VSS Bulk = VSS

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

3.5 FA_MIRROR - details 1/4

24-transistor mirror full-adder core. CON = NOT(AB + C(A+B)); SN = NOT(ABC + CON(A+B+C)). Outputs are inverted carry and sum.

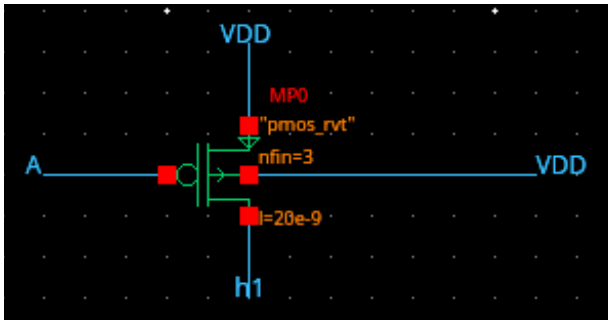
Inputs: A, B, C
Outputs: CON, SN
Supplies: VDD, VSS



Cadence cell symbol

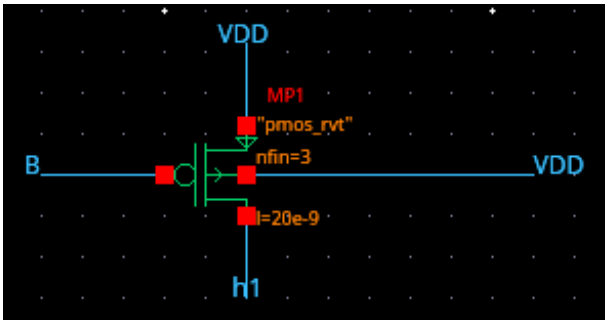
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MP0 | PMOS | 3 fins



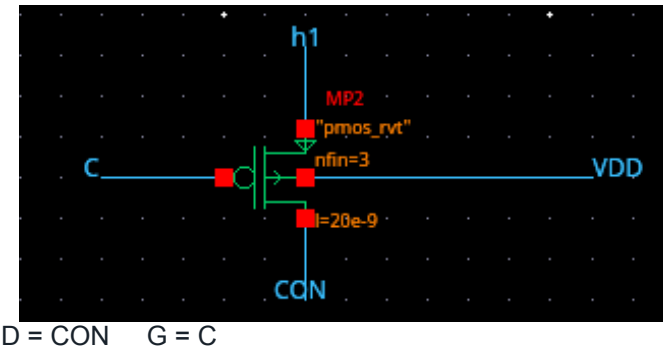
D = h1 G = A
S = VDD Bulk = VDD

MP1 | PMOS | 3 fins



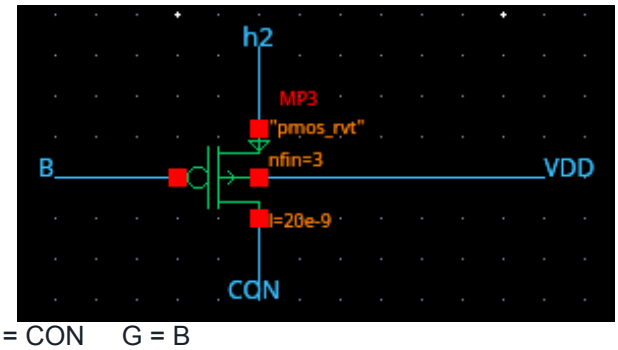
D = h1 G = B
S = VDD Bulk = VDD

MP2 | PMOS | 3 fins



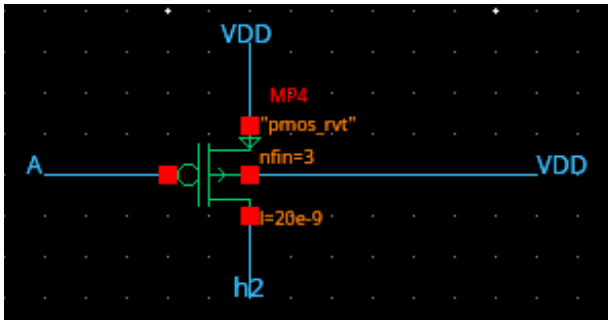
D = CON G = C
S = h1 Bulk = VDD

MP3 | PMOS | 3 fins



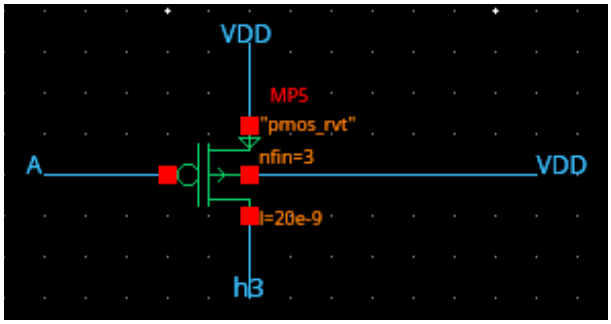
D = CON G = B
S = h2 Bulk = VDD

MP4 | PMOS | 3 fins



D = h2 G = A
S = VDD Bulk = VDD

MP5 | PMOS | 3 fins



D = h3 G = A
S = VDD Bulk = VDD

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

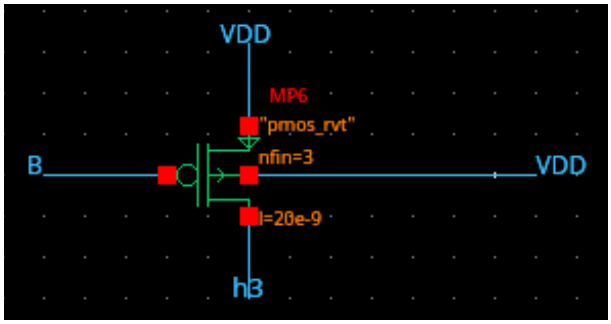
3.5 FA_MIRROR - details 2/4

24-transistor mirror full-adder core. CON = NOT(AB + C(A+B)); SN = NOT(ABC + CON(A+B+C)). Outputs are inverted carry and sum.

Inputs: A, B, C
Outputs: CON, SN
Supplies: VDD, VSS

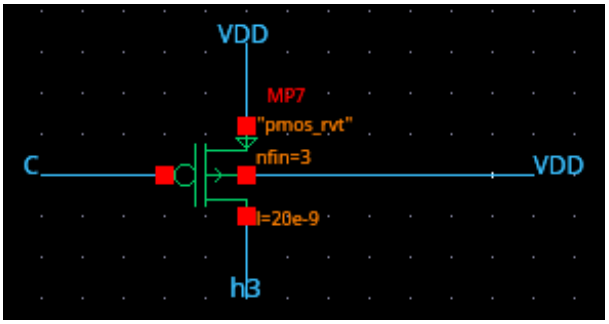
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MP6 | PMOS | 3 fins



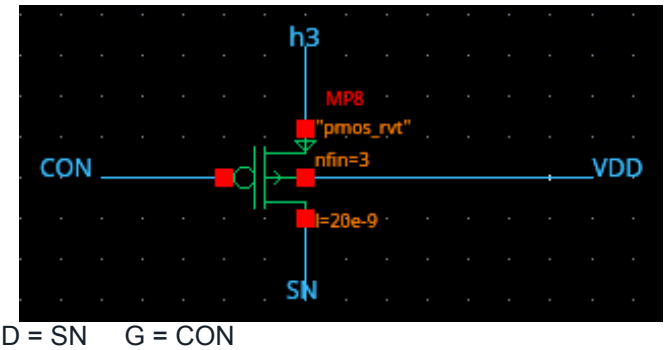
D = h3 G = B
S = VDD Bulk = VDD

MP7 | PMOS | 3 fins



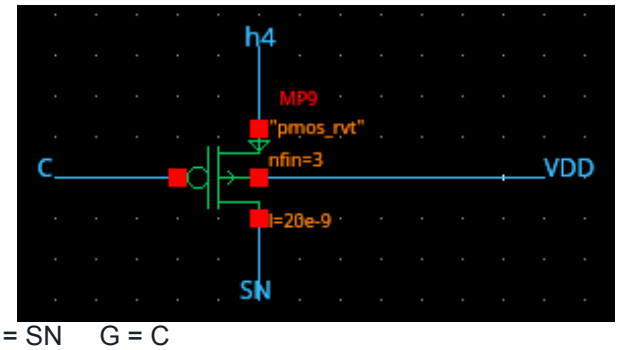
D = h3 G = C
S = VDD Bulk = VDD

MP8 | PMOS | 3 fins



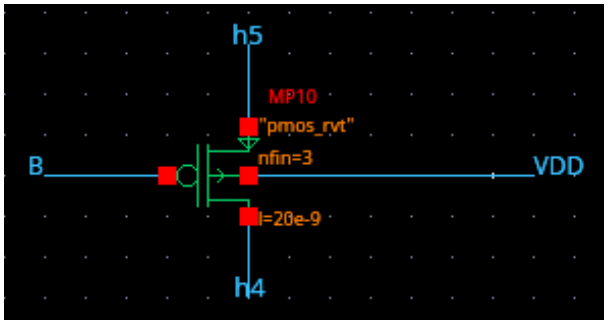
D = SN G = CON
S = h3 Bulk = VDD

MP9 | PMOS | 3 fins



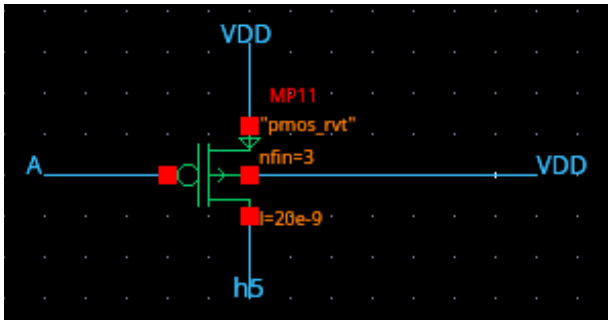
D = SN G = C
S = h4 Bulk = VDD

MP10 | PMOS | 3 fins



D = h4 G = B
S = h5 Bulk = VDD

MP11 | PMOS | 3 fins



D = h5 G = A
S = VDD Bulk = VDD

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

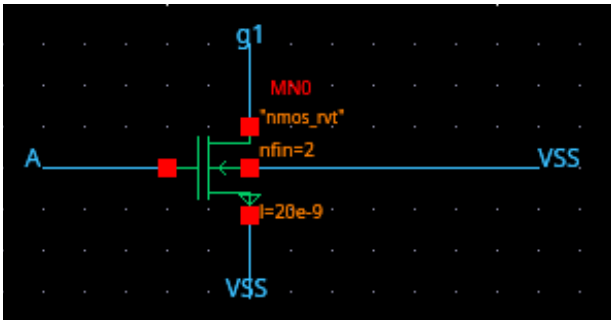
3.5 FA_MIRROR - details 3/4

24-transistor mirror full-adder core. CON = NOT(AB + C(A+B)); SN = NOT(ABC + CON(A+B+C)). Outputs are inverted carry and sum.

Inputs: A, B, C
Outputs: CON, SN
Supplies: VDD, VSS

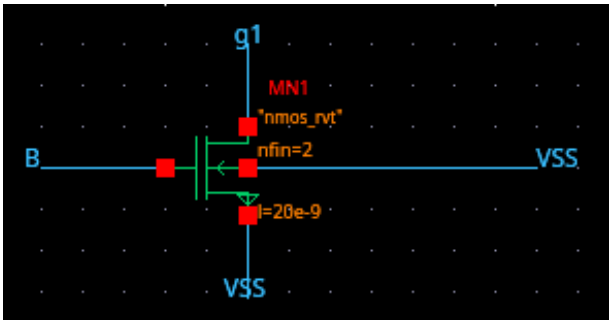
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MN0 | NMOS | 2 fins



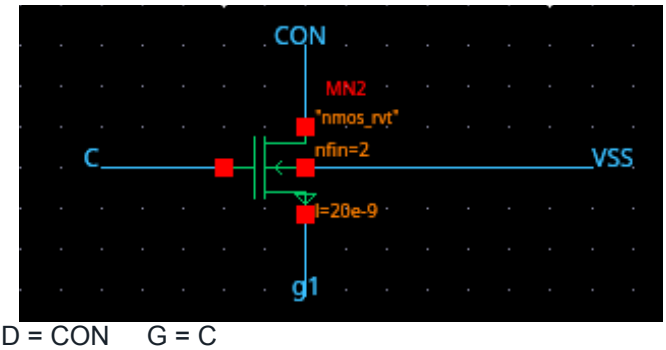
D = g1 G = A
S = VSS Bulk = VSS

MN1 | NMOS | 2 fins



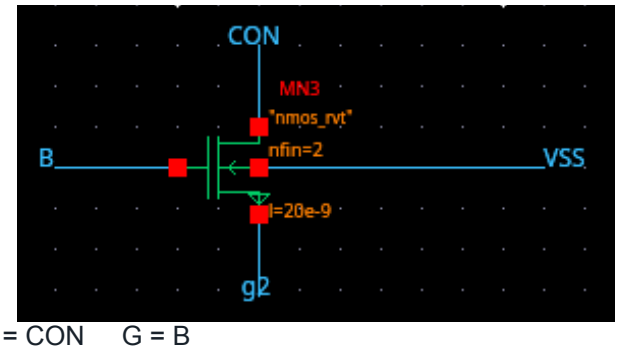
D = g1 G = B
S = VSS Bulk = VSS

MN2 | NMOS | 2 fins



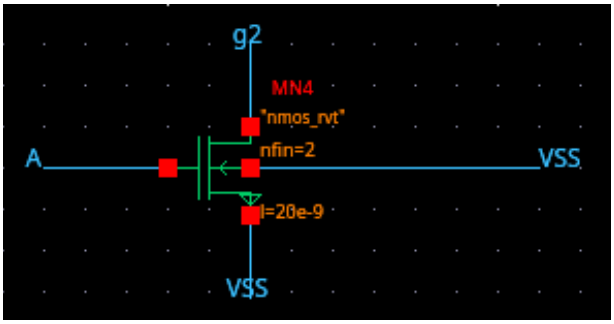
D = CON G = C
S = g1 Bulk = VSS

MN3 | NMOS | 2 fins



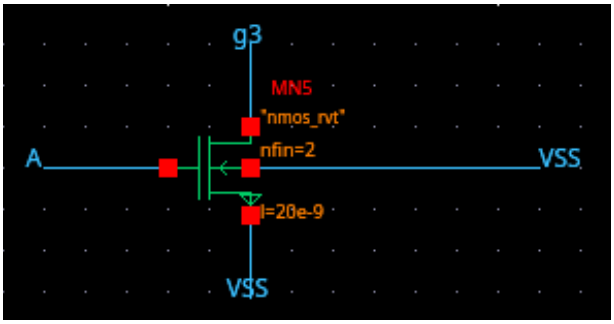
D = CON G = B
S = g2 Bulk = VSS

MN4 | NMOS | 2 fins



D = g2 G = A
S = VSS Bulk = VSS

MN5 | NMOS | 2 fins



D = g3 G = A
S = VSS Bulk = VSS

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

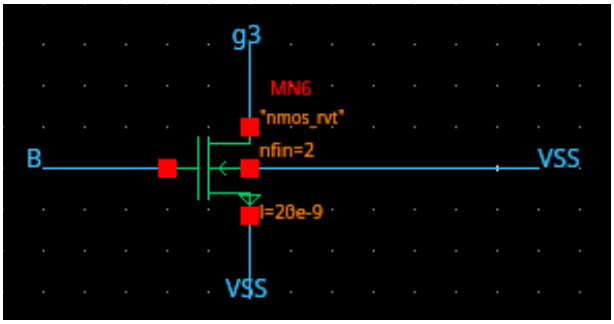
3.5 FA_MIRROR - details 4/4

24-transistor mirror full-adder core. CON = NOT(AB + C(A+B)); SN = NOT(ABC + CON(A+B+C)). Outputs are inverted carry and sum.

Inputs: A, B, C
Outputs: CON, SN
Supplies: VDD, VSS

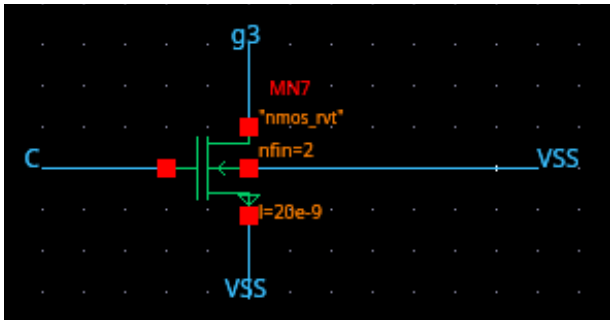
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MN6 | NMOS | 2 fins



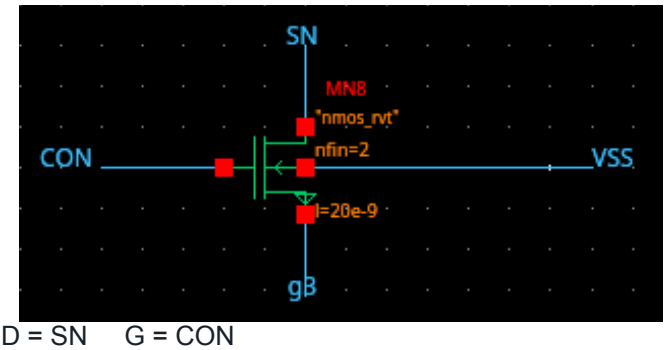
D = g3 G = B
S = VSS Bulk = VSS

MN7 | NMOS | 2 fins



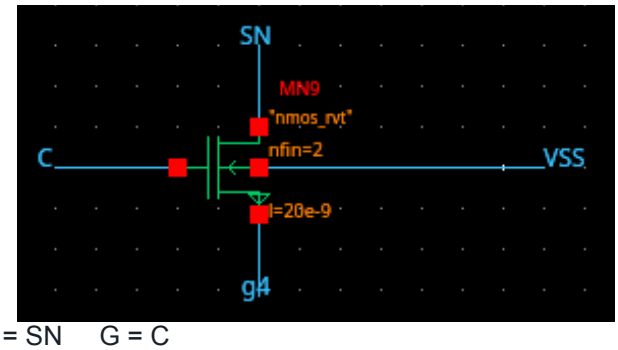
D = g3 G = C
S = VSS Bulk = VSS

MN8 | NMOS | 2 fins



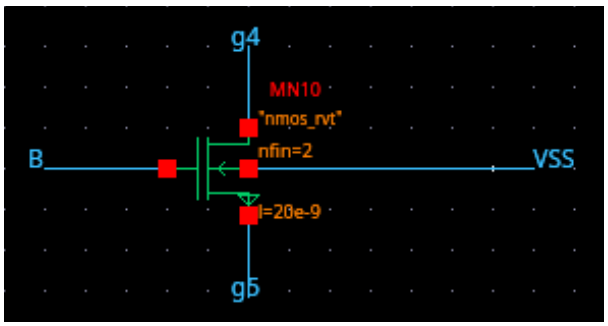
D = SN G = CON
S = g3 Bulk = VSS

MN9 | NMOS | 2 fins



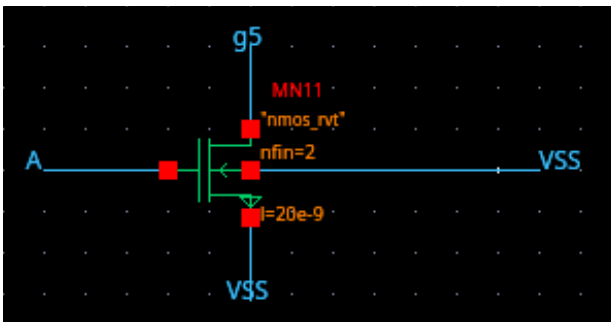
D = SN G = C
S = g4 Bulk = VSS

MN10 | NMOS | 2 fins



D = g4 G = B
S = g5 Bulk = VSS

MN11 | NMOS | 2 fins



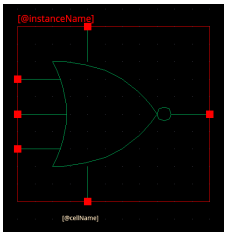
D = g5 G = A
S = VSS Bulk = VSS

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

3.6 NOR3

Three-input NOR: Y = NOT(A OR B OR C). 6 transistors. Produces output Z1 in the ReLU block.

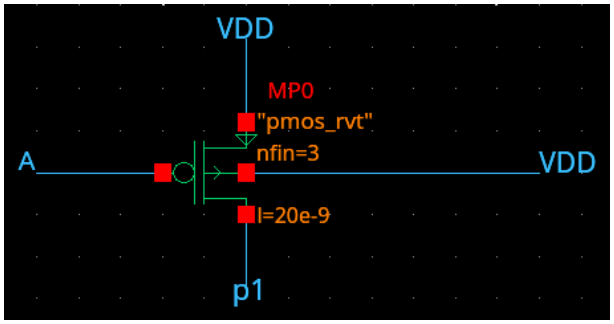
Inputs: A, B, C
Outputs: Y
Supplies: VDD, VSS



Cadence cell symbol

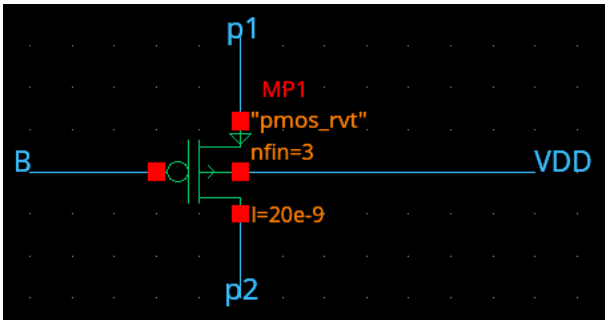
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MP0 | PMOS | 3 fins



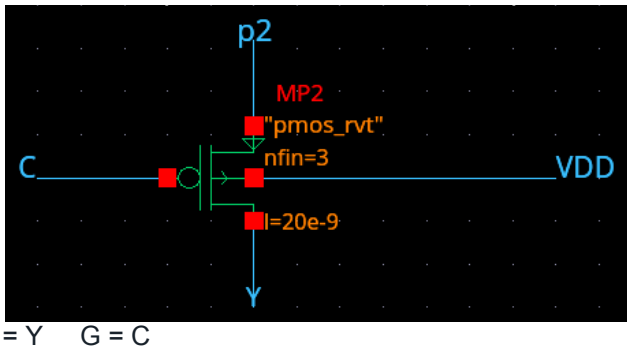
D = p1 G = A
S = VDD Bulk = VDD

MP1 | PMOS | 3 fins



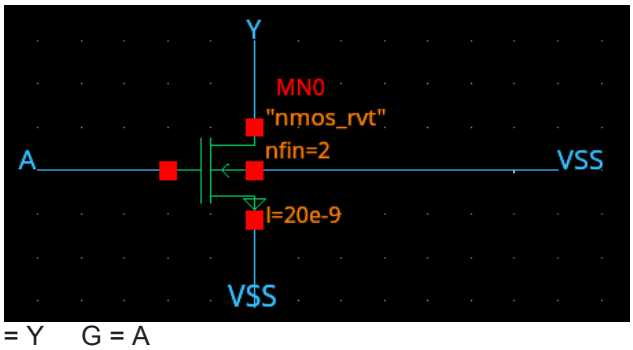
D = p2 G = B
S = p1 Bulk = VDD

MP2 | PMOS | 3 fins



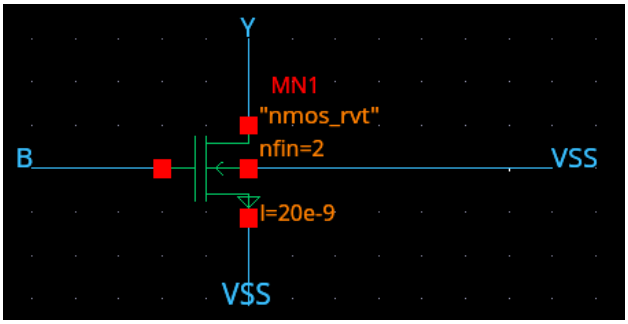
D = Y G = C
S = p2 Bulk = VDD

MN0 | NMOS | 2 fins



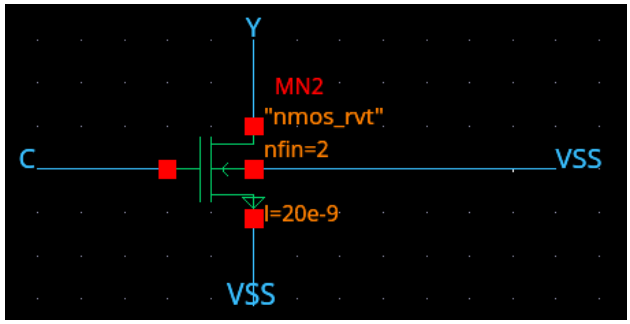
D = Y G = A
S = VSS Bulk = VSS

MN1 | NMOS | 2 fins



D = Y G = B
S = VSS Bulk = VSS

MN2 | NMOS | 2 fins



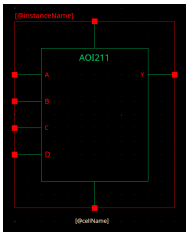
D = Y G = C
S = VSS Bulk = VSS

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

3.7 AOI211 - details 1/2

AND-OR-INVERT: Y = NOT(A*B + C + D). 8 transistors. Produces output Z0 in the ReLU block.

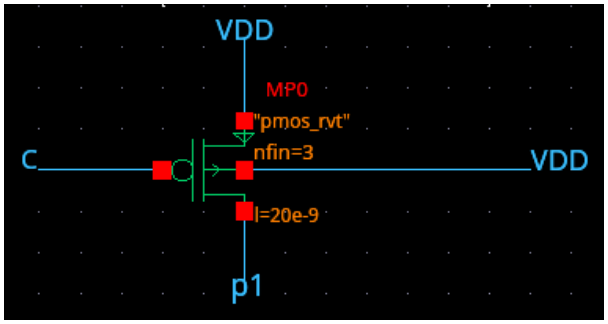
Inputs: A, B, C, D
Outputs: Y
Supplies: VDD, VSS



Cadence cell symbol

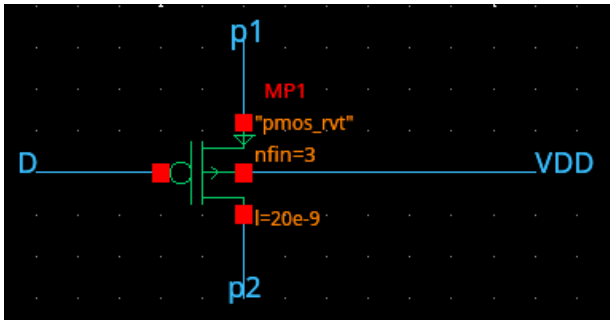
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MP0 | PMOS | 3 fins



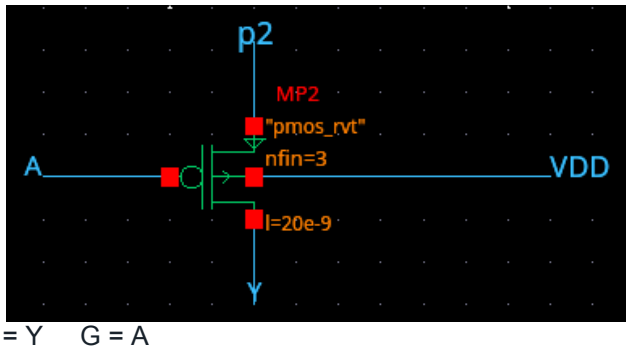
D = p1 G = C
S = VDD Bulk = VDD

MP1 | PMOS | 3 fins



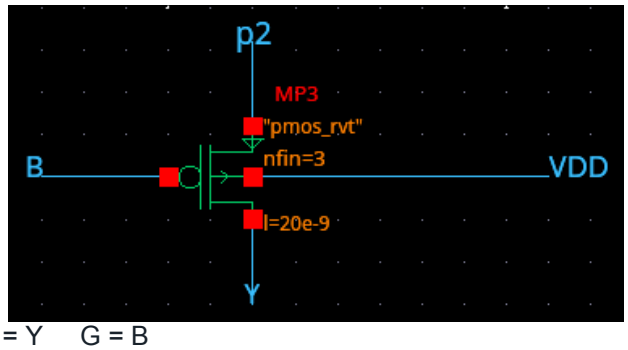
D = p2 G = D
S = p1 Bulk = VDD

MP2 | PMOS | 3 fins



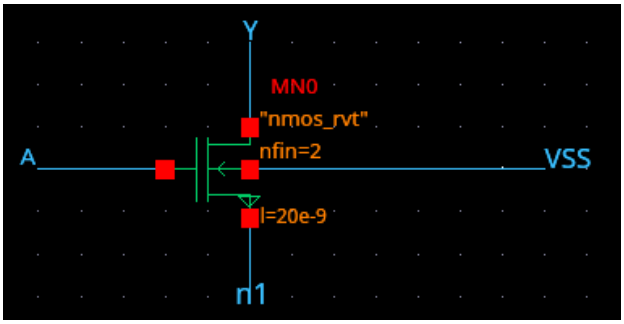
D = Y G = A
S = p2 Bulk = VDD

MP3 | PMOS | 3 fins



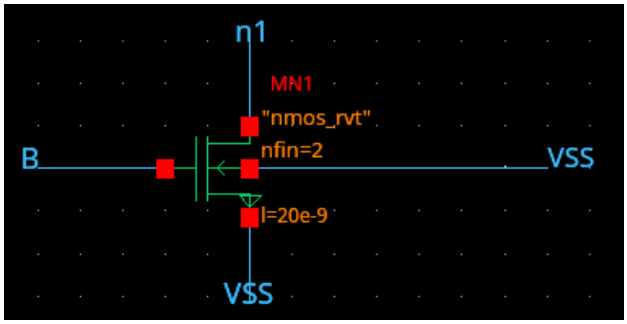
D = Y G = B
S = p2 Bulk = VDD

MN0 | NMOS | 2 fins



D = Y G = A
S = n1 Bulk = VSS

MN1 | NMOS | 2 fins



D = n1 G = B
S = VSS Bulk = VSS

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

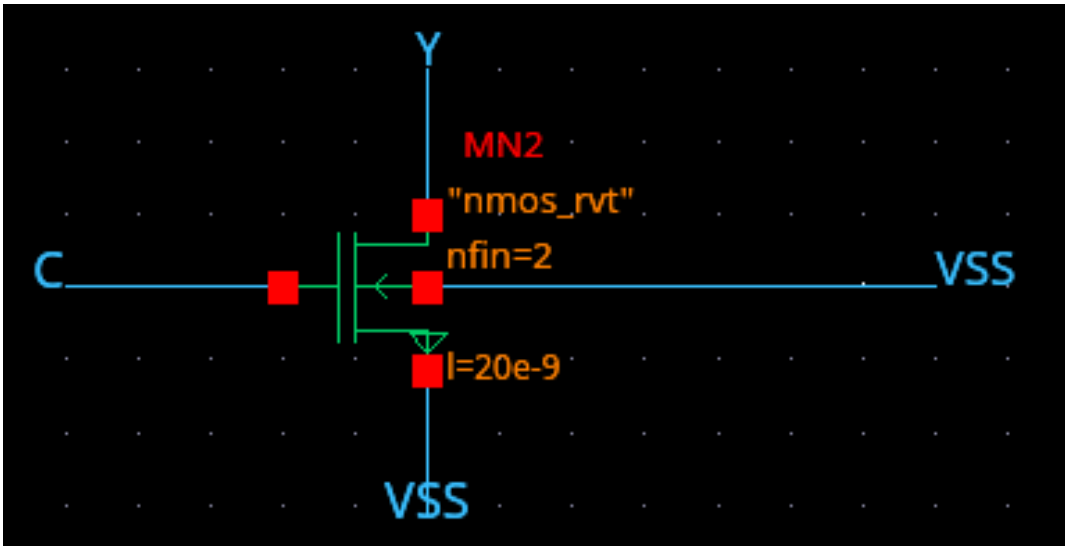
3.7 AOI211 - details 2/2

AND-OR-INVERT: Y = NOT(A*B + C + D). 8 transistors. Produces output Z0 in the ReLU block.

Inputs: A, B, C, D
Outputs: Y
Supplies: VDD, VSS

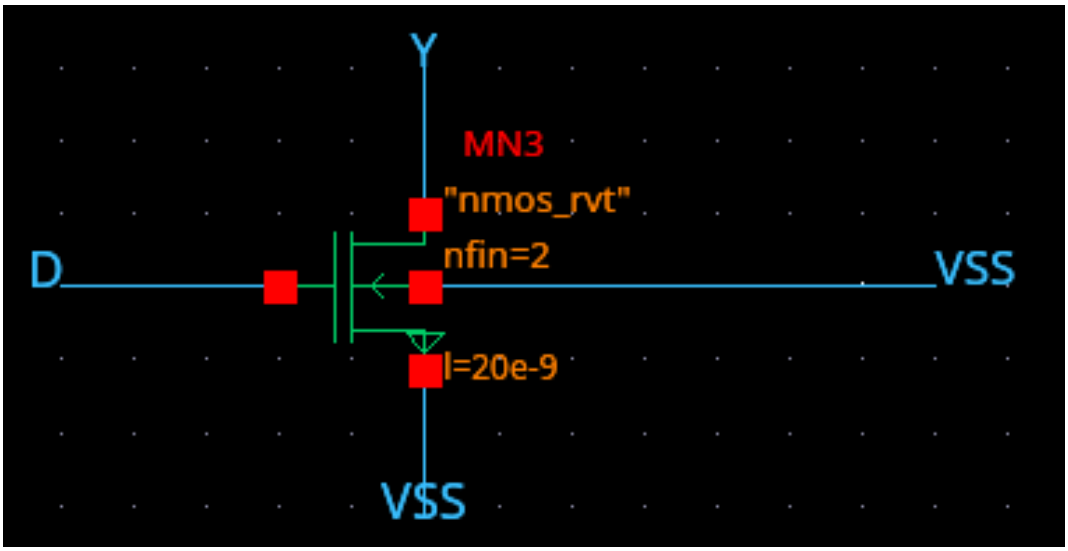
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MN2 | NMOS | 2 fins



D = Y G = C
S = VSS Bulk = VSS

MN3 | NMOS | 2 fins



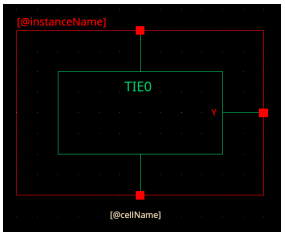
D = Y G = D
S = VSS Bulk = VSS

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

3.8 TIELO

Tie-low cell: Y = 0. 2 transistors. Drives Z2 without connecting the output pin directly to VSS.

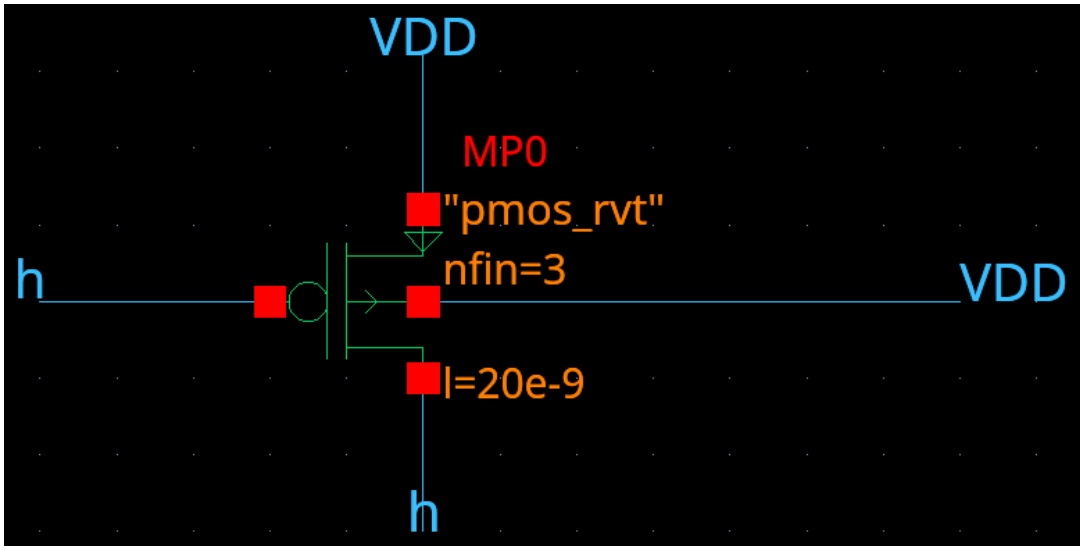
Inputs: None
Outputs: Y
Supplies: VDD, VSS



Cadence cell symbol

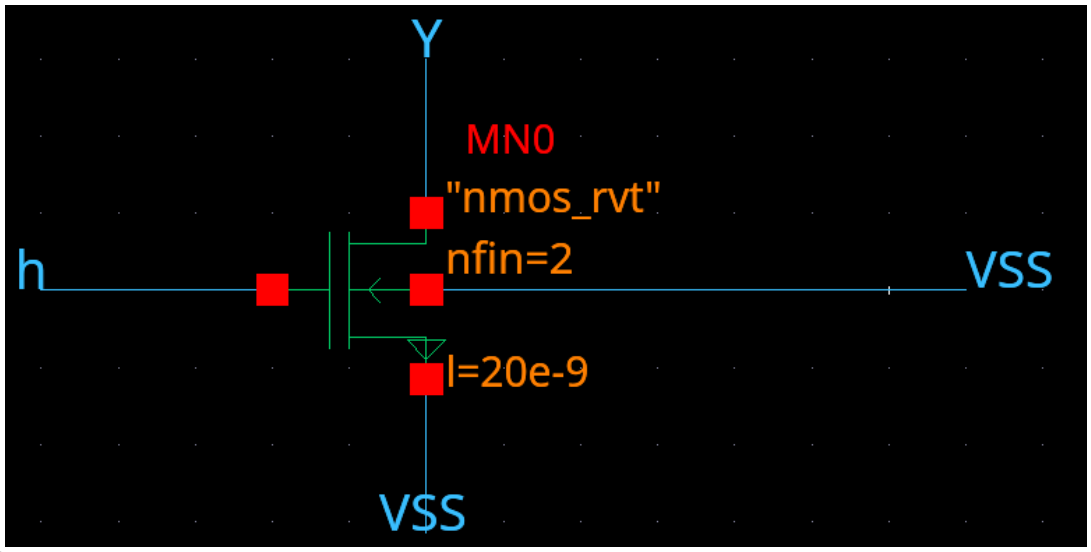
Enlarged panels from the original Cadence schematic. Equal net labels are electrically connected.

MP0 | PMOS | 3 fins



D = h G = h
S = VDD Bulk = VDD

MN0 | NMOS | 2 fins



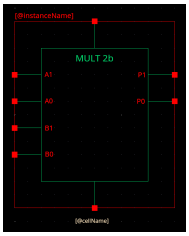
D = Y G = h
S = VSS Bulk = VSS

ASAP7 RVT: L = 20 nm; NMOS W = 54 nm (2 fins); PMOS W = 81 nm (3 fins).

3.9 MULT2 - two-bit multiplier

22 transistors. P0 = A0*B0; P1 = A0*B0*(A1 XOR B1). The sign path reuses the NAND output n0.

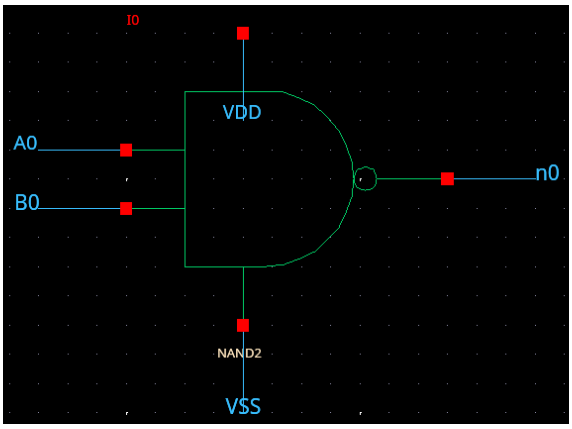
Inputs: A1, A0, B1, B0
Outputs: P1, P0
Supplies: VDD, VSS



Cadence cell symbol

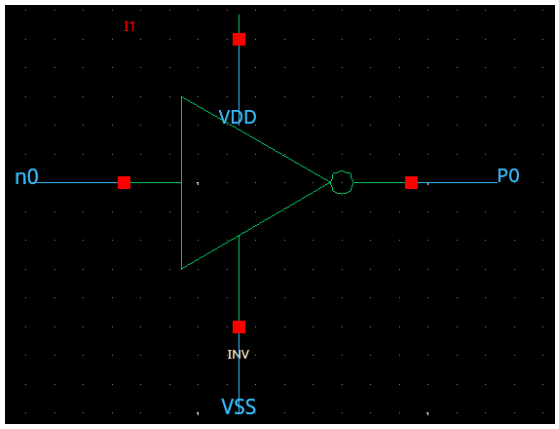
Original Cadence instance details with readable pin-to-net labels. Equal net names connect across panels.

I0 | NAND2



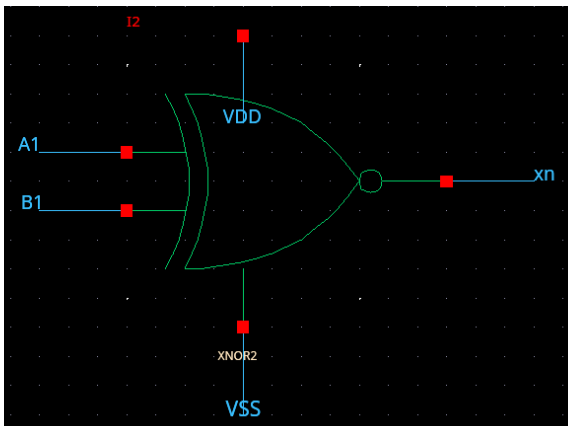
A = A0; B = B0; Y = n0

I1 | INV



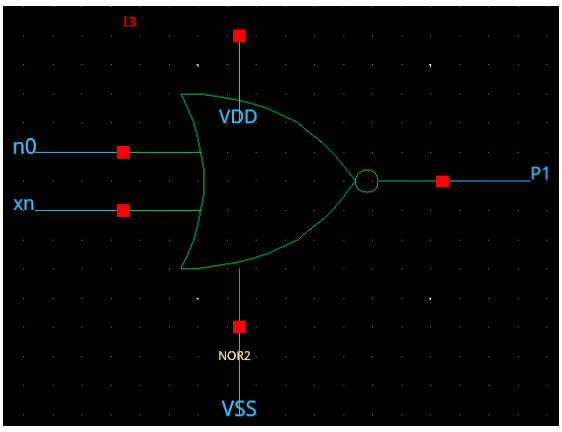
A = n0; Y = P0

I2 | XNOR2



A = A1; B = B1; Y = xn

I3 | NOR2



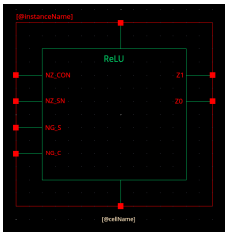
A = n0; B = xn; Y = P1

Every instance has VDD = VDD and VSS = VSS; supplies are explicit inputOutput pins.

3.10 RELU - count-to-output logic

14 transistors. The inverted nonzero-count outputs feed NOR3 and AOI211 directly. The negative count uses true-polarity sum and carry.

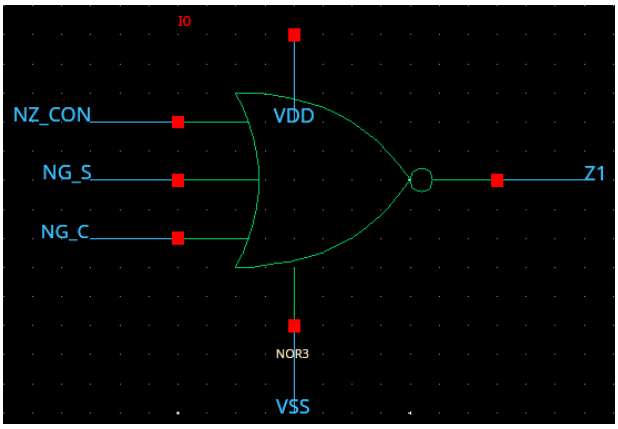
Inputs: NZ_CON, NZ_SN, NG_S, NG_C
Outputs: Z1, Z0
Supplies: VDD, VSS



Cadence cell symbol

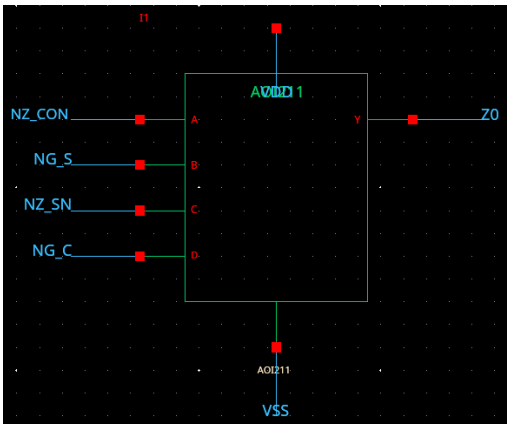
Original Cadence instance details with readable pin-to-net labels. Equal net names connect across panels.

I0 | NOR3



A = NZ_CON; B = NG_S; C = NG_C; Y = Z1

I1 | AOI211



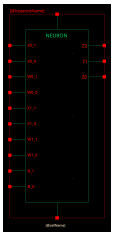
A = NZ_CON; B = NG_S; C = NZ_SN; D = NG_C; Y = Z0

Every instance has VDD = VDD and VSS = VSS; supplies are explicit inputOutput pins.

4. Top-level NEURON - details 1/2

112 transistors. Two multipliers, two mirror full adders, two inverters, the ReLU block, and TIELO compute Z = max(0, x0*w0 + x1*w1 + b).

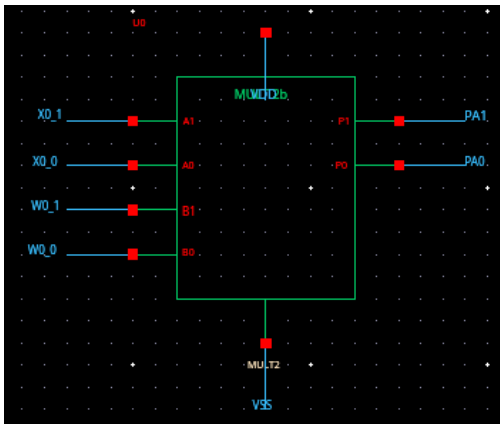
Inputs: X0_1, X0_0, W0_1, W0_0, X1_1, X1_0, W1_1, W1_0, B_1, B_0
Outputs: Z2, Z1, Z0
Supplies: VDD, VSS



Cadence cell symbol

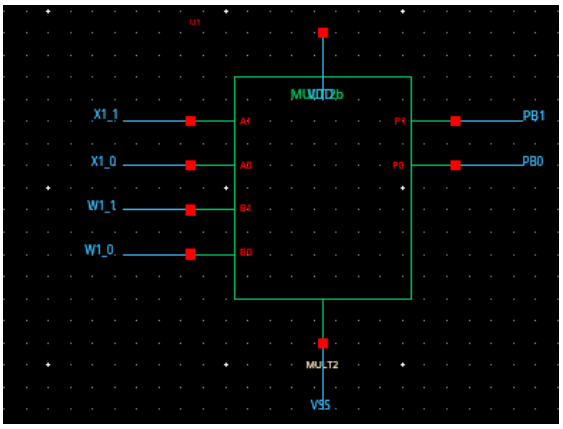
Original Cadence instance details with readable pin-to-net labels. Equal net names connect across panels.

U0 | MULT2



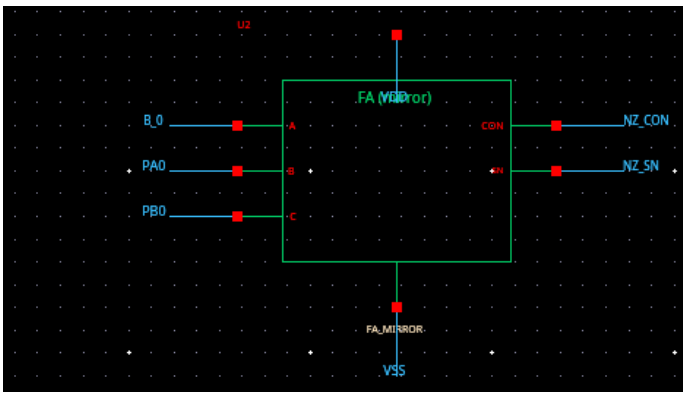
A1 = X0_1; A0 = X0_0; B1 = W0_1; B0 = W0_0; P1 = PA1; P0 = PA0

U1 | MULT2



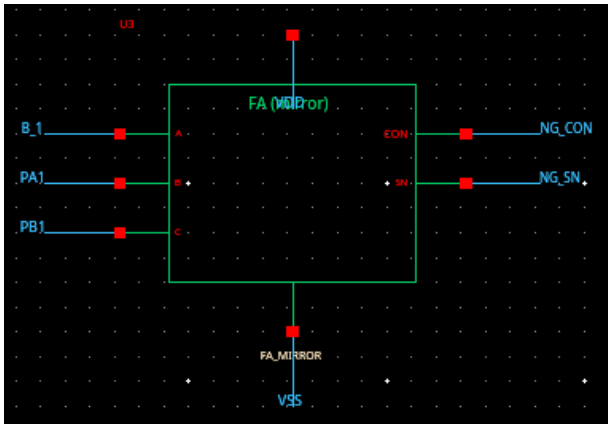
A1 = X1_1; A0 = X1_0; B1 = W1_1; B0 = W1_0; P1 = PB1; P0 = PB0

U2 | FA_MIRROR



A = B_0; B = PA0; C = PB0; CON = NZ_CON; SN = NZ_SN

U3 | FA_MIRROR



A = B_1; B = PA1; C = PB1; CON = NG_CON; SN = NG_SN

Every instance has VDD = VDD and VSS = VSS; supplies are explicit inputOutput pins.

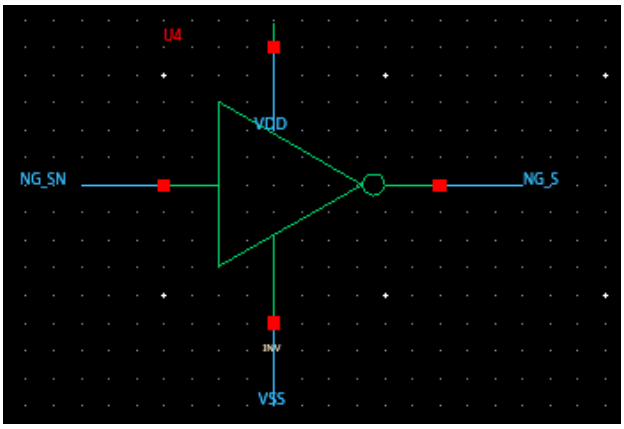
4. Top-level NEURON - details 2/2

112 transistors. Two multipliers, two mirror full adders, two inverters, the ReLU block, and TIELO compute Z = max(0, x0*w0 + x1*w1 + b).

Inputs: X0_1, X0_0, W0_1, W0_0, X1_1, X1_0, W1_1, W1_0, B_1, B_0
Outputs: Z2, Z1, Z0
Supplies: VDD, VSS

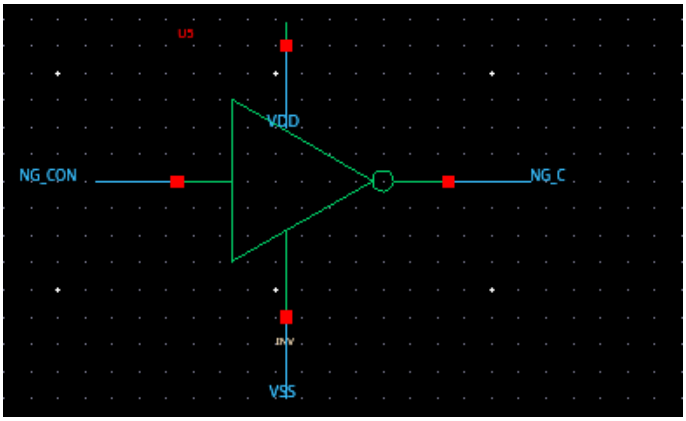
Original Cadence instance details with readable pin-to-net labels. Equal net names connect across panels.

U4 | INV



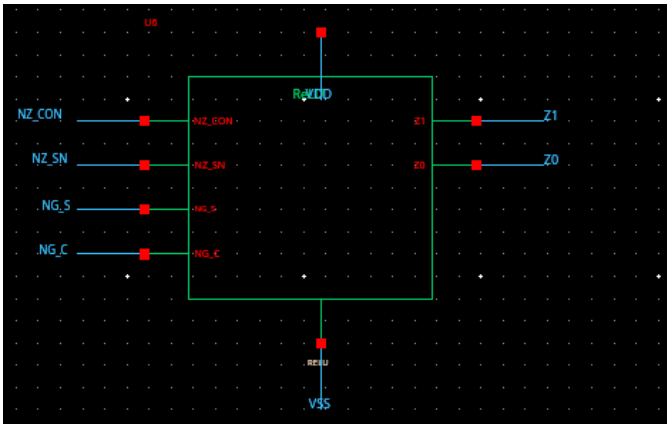
A = NG_SN; Y = NG_S

U5 | INV



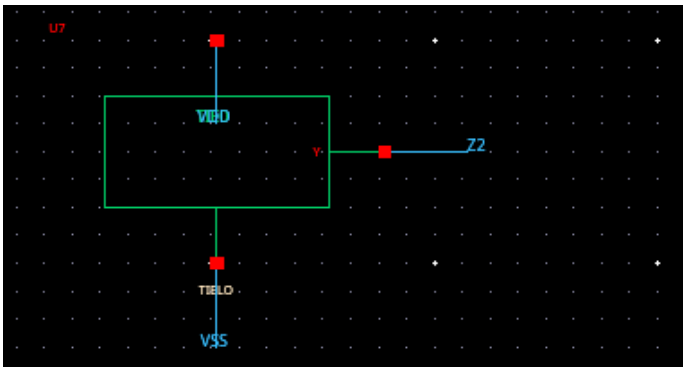
A = NG_CON; Y = NG_C

U6 | RELU



NZ_CON = NZ_CON; NZ_SN = NZ_SN; NG_S = NG_S; NG_C = NG_C; Z1 = Z1; Z0 = Z0

U7 | TIELO



Y = Z2

Every instance has VDD = VDD and VSS = VSS; supplies are explicit inputOutput pins.

4. Neuron summary and verification

The top-level NEURON contains 112 transistors. The preceding pages show all eight top-level instances and every constituent cell.

Block	Instances	Transistors each	Total
MULT2	2	22	44
FA_MIRROR	2	24	48
INV	2	2	4
RELU	1	14	14
TIELO	1	2	2
NEURON			**112**

Critical path (estimated). x1/w1 to XNOR2 (inverter + AOI22), NOR2, FA_MIRROR (C input), INV, NOR3/AOI211, then Z. The static bias weight drives input A; the late product bits drive C.

5. Functional verification

MS1 does not require simulations. The following checks were recorded for the design before layout.

Switch-level check. Every leaf-gate input and all 243 legal (x0, w0, x1, w1, b) combinations were checked. All 243 produce Z = max(0, y). The saved Cadence connectivity was checked against the source netlists.

Pre-layout HSPICE check. The NEURON schematic was netlisted from Virtuoso (hspiceD) and simulated with ASAP7 TT models at VDD = 0.9 V, 25 degrees C, and 1 fF per output. The testbench applies all 243 legal combinations for 400 ps each and samples Z2 Z1 Z0 at the end of each step. All 243 cases match max(0, y). Sampled logic-1 levels were at least 0.900 V; sampled logic-0 levels were at most 34 microvolts.